

RWANDA TVET BOARD
COMPETENCY-BASED ASSESSMENT
[FORMATIVE ASSESSMENT]

Program: SOFTWARE DEVELOPMENT | **Level:** 5 | **Total Marks:** 30

Module: Machine Learning Application | **Date:** 2026-05-24 | **Time:** 1 Hour 30 Minutes

Name: _____ Index No: _____ Class: _____

SECTION A: TRUE / FALSE

Answer True (T) or False (F).

- 1 Understanding test is required before applying Data Pre-processing. (1 marks)

.

Answer: **True** **False**

- 2 Understanding deploy is required before applying Data Pre-processing. (1 marks)

.

Answer: **True** **False**

- 3 It is possible to complete applying Data Pre-processing without using build. (1 marks)

.

Answer: **True** **False**

- 4 It is possible to complete applying Data Pre-processing without using explain. (1 marks)

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Answer: **True** **False**

SECTION B: MULTIPLE CHOICE

Choose ONE correct answer and circle the letter.

- 5 Which approach to learning is most effective when applying Data Pre-processing? (1 marks)

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- A. Model deployment
- B. o Training an RNN
- C. Model integration in machine learning
- D. learning

Answer: _____

- 6 What is the main purpose of apply when applying Data Pre-processing? (1 marks)

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- A. Machine learning
- B. apply
- C. Random Forest
- D. Model integration in machine learning

Answer: _____

- 7 Which design using manage would best support applying Data Pre-processing? (1 marks)

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- A. Initializing a model
- B. Communicator
- C. manage
- D. Consistent data

Answer: _____

- 8 Which of the following correctly defines identify? (1 marks)

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- A. Big data

- B. identify
 - C. Advantage
 - D. Fitting a model
- Answer: _____

9 How would you correctly apply Machine learning life cycle when applying Data Pre-processing? (1 marks)

- A. Data analysis
 - B. Data wrangling
 - C. Data Cleaning
 - D. Machine learning life cycle
- Answer: _____

1 What is the key difference when using Data preparation for applying Data Pre-processing? (1 marks)

- A. Data preparation
 - B. Advantage
 - C. Model integration in machine learning
 - D. Random Forest
- Answer: _____

1 How would you correctly apply Data wrangling when applying Data Pre-processing? (1 marks)

- A. Mietech LTD
 - B. Model integration
 - C. Example Plant growth
 - D. Data wrangling
- Answer: _____

SECTION C: MATCHING

Match each term in Column A with Column B.

1 Match each term in Column A with its correct description in Column B. (2 marks)

Column A (Terms)	Column B (Descriptions)
1. Machine learning life cycle	A. is a cyclic process to build an efficient machine learning project.
2. Data Gathering	B. is the first step of the machine learning life cycle. The goal of this step is to
3. Data preparation	C. is a step where we put our data into a suitable place and prepare it to
4. Data wrangling	D. is the process of cleaning and converting raw data into a useable format. It

Answers: 1.____ 2.____ 3.____ 4.____

1 Match each term in Column A with its correct description in Column B. (2 marks)

Column A (Terms)	Column B (Descriptions)
1. Machine learning life cycle	A. is a cyclic process to build an efficient machine learning project.
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4. Data wrangling	D. is the process of cleaning and converting raw data into a useable format. It

Answers: 1.____ 2.____ 3.____ 4.____

SECTION D: OPEN-ENDED

Answer ALL questions. Write in the spaces provided.

1 Explain how analyze works in the context of applying Data Pre-processing. **(-3 marks)**

4

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1 Explain how design works in the context of applying Data Pre-processing. **(3 marks)**

5

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1 Explain how develop works in the context of applying Data Pre-processing. **(3 marks)**

6

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1 Demonstrate how to use install when applying Data Pre-processing. Provide a **(3 marks)**
7 step-by-step explanation.

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1 Define create and explain its role in applying Data Pre-processing. **(3 marks)**

8

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1 Evaluate the effectiveness of describe when applying Data Pre-processing. Justify with **(3 marks)**
9 examples.

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2 Analyze the importance of Data Gathering in applying Data Pre-processing. Identify **(3 marks)**
0 advantages and limitations.

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END OF EXAMINATION | Formative | Machine Learning Application | Total: 30 marks